



GhostNet, ShuffleNet, MobileNetV2 and ShuffleNetV2 with the proposed CMDA module for better performance and real-time video action recognition.

The authors aim to solve two problems with video action recognition: (1) the high computation and storage overhead and (2) the challenge of adapting to temporal and spatial information in the Fast pathways and Slow pathways. These are important aspects to address because they can impact the efficiency and accuracy of the video action recognition system.

If this problem can indeed be solved, it would benefit many people engaged in video action recognition. It would help to improve the accuracy and speed of the performance of video action recognition tasks, making the task more efficient and cost-effective. This would have a positive impact not only on research but also on current practices in the field. For instance, it could lead to improved video surveillance systems, more reliable automated driving systems, and higher accuracy in video recognition tasks. Additionally, it could facilitate real-time video action recognition, which would be beneficial in many other fields such as gaming, artificial intelligence, and so on.

This can be of great help in terms of saving computation and storage costs for resource-limited applications. We can see that all the proposed two-stream architectures outperform both SlowFast and their corresponding lightweight architectures. This verifies the effectiveness of the CMDA module in explicitly transferring information and concentrating on the critical area of the video. Moreover, most of the proposed architectures achieve both better accuracy and smaller FLOPs numbers than the original SlowFast model, which means that the authors' proposed architectures are more efficient.

on well-known 2D efficient architectures (ShuffleNet, ShuffleNetV2, MobileNet, MobileNetV2, and GhostNet) which consist of a single stream operating at two different framerates and exchange information bidirectionally with the help of the CMDA fusion module.

These methods were chosen because they are efficient and effective, providing sufficient and explicit information exchange to improve the representation of interests while being fast and resource efficient. The strengths of these methods include the use of attention mechanisms to selectively focus on salient parts and areas to capture temporal and visual structure better, as well as a delicate procedure designed to execute heavy calculations in fewer parameters. Additionally, different methods are used to fuse information between the Slow and Fast pathways, such as downsampling and convolution layers, allowing for more efficient utilization of resources. The weaknesses of these methods include the potential to miss some keyframes when downsampling, and the use of a single convolution layer which may not be sufficient to exchange enough information.

The authors made a lot of good choices in their methods. They used attention mechanisms such as spatial-temporal attention and efficient channel attention to exchange information between the Slow and Fast pathways and designed a fusion module to reduce model complexity and introduce residual connections for better learning. Additionally, they used 3D self-attention and a 1D Conv with an adaptive kernel size to capture local cross-channel interaction efficiently and implemented a bottleneck design to reduce computation and optimize execution order.

Other methods that may have worked better include using a Gated Recurrent Unit (GRU) or Long Short-Term Memory (LSTM) layer to process temporal data more effectively. Try different temporal pooling strategies: Temporal pooling is used to reduce the temporal dimension of the feature maps for better performance. Different strategies like max pooling, average pooling and adaptive pooling can be tried to determine the best pooling strategy.

3. Results

The experimental results presented in the paper show that the proposed CMDA module improves the accuracy of SlowFast networks on the Kinetics-400 and 20BN-Jester-v1 datasets. The evaluation strategy used in the paper involves comparing the proposed CMDA fusion module to the original unidirectional fusion module and the recent state-of-the-art models. The main findings of the paper indicate that the proposed CMDA module can achieve better performance with fewer parameters and fewer FLOPs than the original unidirectional fusion module, as well as higher accuracy than the recent state-of-the-art lightweight models. Furthermore, the paper also provides visualisations of the salient feature maps using Grad-CAM, which demonstrate the effectiveness of the CMDA fusion module in improving the accuracy of the SlowFast networks.

The assessment strategies used in this paper include quantitative analysis, qualitative analysis, and runtime performance analysis. For the quantitative analysis, the authors applied the proposed CMDA fusion module to various classical efficient networks and compared the results with the SlowFast network as well as other state-of-the-art lightweight action recognition methods on two datasets, Kinetics-400 and 20BN-Jester-v1. The results were evaluated in terms of accuracy, number of parameters, number of floating point operations, and frames per second. For the qualitative analysis, the authors used Grad-CAM to visualize the salient feature maps for the SlowFast networks and demonstrated the improved ability of focus and sense perception with the help of the bidirectional information exchange paths. Additionally, the authors did not do any ablation studies to further show the effectiveness of their proposed models.

The findings from these experiments could be persuasive for potential users to adopt the proposed method as it has achieved better performance in accuracy, with fewer parameters and FLOPs than the original SlowFast networks. The visualizations in Fig. 1 also provide an insight into how attention mechanisms and bidirectional information-exchanging paths enhance the performance of the Slow and Fast pathways.

To make the case even more convincing, other experiments that could have been conducted include testing the proposed method on different datasets, such as the UCF-101 and HMDB-51 datasets, as well as testing its performance in different scenarios, such as in low-light conditions or scenes

with challenging backgrounds. In addition, measuring the time complexity of the proposed method and comparing it to the original SlowFast network would provide further support.

4. Conclusions

Strength: 1. Presents a clear and concise overview of the current state of video action recognition and the challenges involved. 2. Presents a comprehensive overview of current state-of-the-art techniques in the field of video action recognition. 3. Introduces an innovative two-stream model for action recognition which can be used for various applications.

Weaknesses: 1. The paper does not provide comprehensive evaluations of the practical applications of the proposed models. 2. Does not provide a comprehensive discussion of the limitations of the proposed models. 3. No intuitive comparison with models with large computational volumes and heavy storage consumption

Despite the encouraging results, several technical issues still need to be addressed before the problem of video action recognition can be considered solved. First, further improvements can be made to the cross-modality dual attention fusion module. For example, exploring more powerful attention mechanisms such as spatial-temporal-channel attention and adaptive kernel size in channel-wise attention blocks can help further boost performance. Second, due to the real-time requirements of many video action recognition applications, more research is needed to accelerate the model inference speed, such as learning efficient computation strategies and exploring more advanced network architectures. Finally, the current datasets are limited in complexity and diversity(For example, in the case of continuous action), so increasing the availability of large-scale, highly-diverse datasets is necessary to further the development of video action recognition.

Suggestions for future research include exploring more efficient fusion strategies, applying the proposed methods to more complex datasets, and conducting more ablation studies to assess the effectiveness of the proposed methods. Additionally, further research could be conducted to explore the potential applications of the proposed methods in other video-related tasks such as video segmentation or video generation.

References

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